

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – _____ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026

Proposed Solution · one free phone call that turns a person's skills into the right work

Why this needs fixing SIH26097 · Ministry of Social Justice & Empowerment
PM-AJAY pays for **skill training** and **small-business support** (up to ₹50,000) for poor Scheduled Caste families. But the course for each person is chosen through paper forms and field officers — so people who cannot read, do not know the modern trades, or speak only a dialect land in the wrong course, drop out, and get no job.

41%

of trained people got a job
— CAG audit of PMKVY,
2025

34%

of SC adults cannot read —
Census 2011

36%

of women have no phone of
their own — NFHS-6, 2023–
24

ONE CALL, SIX STEPS — ABOUT 10 MINUTES, ON ANY PHONE



1

Missed call → call-back

The family gives a free missed call. SkillCall calls back within a minute from an official 1600 govt number, in the language they pick on the keypad.



2

Facts by keypad

Age, education, travel distance, job-or-own-business — pressed on keys. A speech mistake can never change who is eligible.



3

He talks, AI listens

"Aap kya kaam karte ho?" The AI maps his answer to an official NCO-2015 occupation, then asks follow-up questions about his skills.



4

Read back & saved

Everything is read back; he presses 1 to confirm or 2 to correct. The profile is stored as one clean table row — not a paper form.



5

1-4 options that fit him

Courses, schemes or jobs near him — only the ones that truly fit. Every fact is checked against govt data before it is spoken. He can ask any doubt.



6

Choose → CSC → officer

He presses 1-4. SMS with Case ID + nearest CSC; documents and e-KYC there; the officer approves in the AJAY app. Follow-up calls at 1, 3, 6 months.

WHAT ONLY SKILLCALL DOES — BEYOND THE SIX STEPS



Smart questions

Each next question depends on the last answer. A bike mechanic is asked about engine repair, tyre change and oil service — not a generic form.



Only government data

The AI answers only from data we load: scheme rules, NQR courses, NCO jobs, training centres. No internet search, nothing invented.



Validator AI — SATYA

Before any fact is spoken, a second checker matches it to a database row. No match → that sentence is dropped, never said.



Cheapest AI first

Free or small models by default. A bigger model only if the small one is unsure or the caller is unhappy; a human calls back if still unresolved.



Web portal too

Log in with mobile number + OTP. The person, the CSC or the officer can view and edit the same profile anytime — or do the whole process on the web.

Also answers the 5 GIA issues named in the brief → planning · beneficiary selection · placement after training · coordination · too few ground staff

TECH STACK

Calls & channels

WhatsApp Business API	Exotel / Plivo • calls + SMS
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Voice engine & AI models

Pipetac voice server	Saaras v3 • speech → text
Bulbul v3 • text → speech	Sarvam-30B / 105B • LLM
IndicConformer (free STT)	Bhashini • MeitY
SATYA validator (ours)	Silero VAD • voice detection

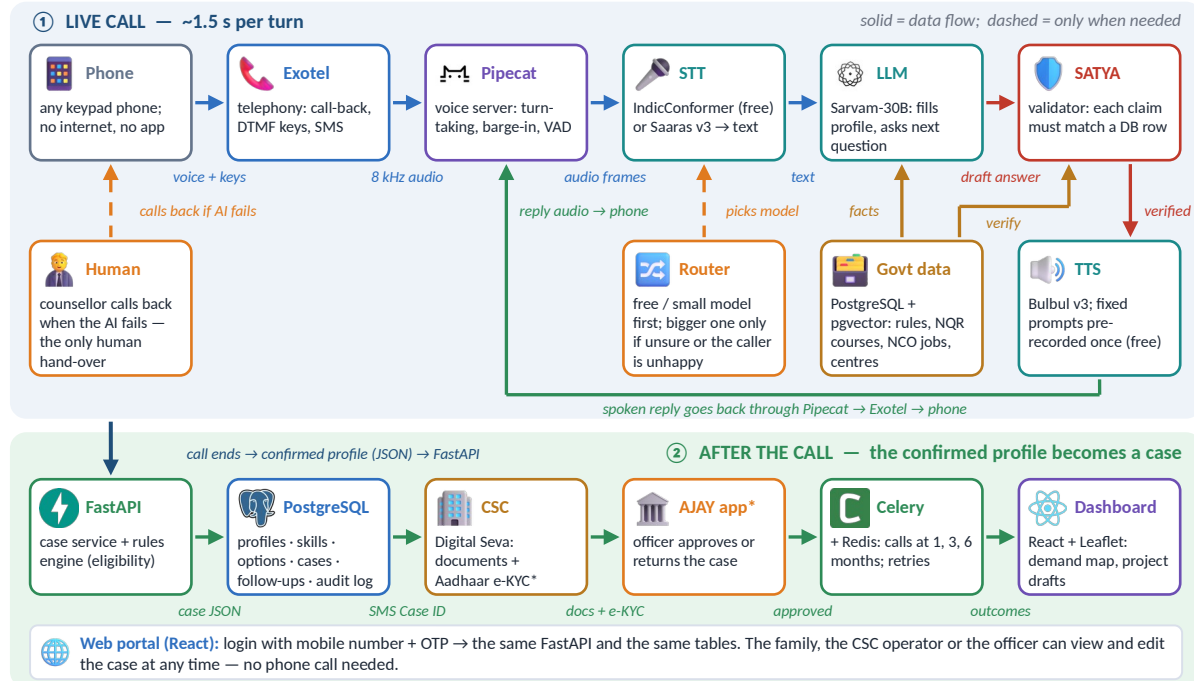
Backend & data

Python	FastAPI
Celery	Redis
PostgreSQL	pgvector • semantic search
PostGIS • distance to centres	JWT auth + audit log

Apps, hosting & govt links

React portal + dashboard	Leaflet maps
Docker	MeghRaj govt cloud*
NCO-2015 • occupation codes	NQR / NCVET • courses
DigiLocker* • documents	AJAY app* • officer workflow

HOW IT WORKS — ARCHITECTURE



AI understands → verified data decides → SATYA checks → humans approve

Data rules: no Aadhaar number stored; raw audio deleted after confirmation; consent logged (DPDP Rules 2025); every edit in the audit log.

* needs a government MoU.

Full architecture, API docs and code → [GitHub repo \(add link\)](#)

Can it be built and run? Yes — every piece already works in India today, every model is available by API now, and one completed call costs about ₹37–83.

ALREADY WORKING IN INDIA — NOTHING HERE IS NEW TECHNOLOGY



10 of 22 languages

UIDAI already runs Sarvam's voice AI for Aadhaar in 10 of its 22 Indian languages, on its own servers.

PIB / MeitY 2025



12 lakh calls a day

Kilkari, the government's health-advice call system, already handles this volume every day.

BMJ Global Health 2021



≤ 20% word error

Best Indian speech AI measured on real phone calls: 36,691 speakers, 15 languages.

Voice of India benchmark 2026



All parts available today

Sarvam APIs, Pipecat, Exotel trial, PostgreSQL, React — the prototype can be built right now.

our prototype plan

WHAT ONE COMPLETED CALL COSTS — ₹ PER PERSON

A completed call = 10-min interview + options + 4 follow-up calls (= 18 min of phone time).

Cost item	How we calculate it	Low	High
Phone line (Exotel)	18 min × ₹0.80–1.80 per minute	₹16.5	₹37.6
Speech → text (Saaras v3)	₹30 per hour of audio	₹9.4	₹9.5
Text → speech (Bulbul v3)	₹3 per 1,000 characters, spoken sentences only	₹2.4	₹6.0
AI brain (Sarvam-30B / 105B)	≤ ₹73 per 1M tokens; ~50–100K tokens per call	₹1.4	₹3.1
SMS + human help + CSC fee	4–6 SMS; 10–20% need a counsellor; 20–40% need CSC help	₹6.6	₹23.5
Hosting & helpline number	servers, database, number — spread per call	₹1.0	₹3.0
TOTAL per completed profile	1,000 missed calls → 517 finished profiles	≈ ₹37	≈ ₹83

RISKS AND HOW WE HANDLE THEM

Village dialects are hard → facts by keypad · every answer read back · bigger model, then a human

People ignore unknown numbers → he calls first; the call-back comes from an official 1600 number

Government links need permission → works standalone today; AJAY app, DigiLocker, EPFO plug in after an MoU

CAN THE GOVERNMENT AFFORD IT?



0.04–0.08% of the yearly budget

Running SkillCall for a whole state costs ₹0.9–1.7 crore a year. PM-AJAY's budget is ₹2,140 crore a year.



₹9–13 lakh to start

A 6-month pilot in one block (about 1,000 families): build, test with real users, then scale to the state.



Nothing new to build on the ground

6.5 lakh CSC centres and the AJAY officer app already exist. SkillCall plugs into them; it does not replace them.

IMPACT AND BENEFITS

Who gains? Poor SC families, CSC operators, district officers and the Ministry. Every number below shows how it was worked out.

WHAT CHANGES



More trained people get jobs

41% → 70%

Only 41% of trained people get a job today; 70% is the government's own target. Per 1,000 trainees, reaching even 50% means 90 more people working.

CAG 2025 · MSDE norms



More people stay in the job

+11 percentage points

e.g. from 41% to 52% still in the job — when a trainee is told the real pay and place before joining (field trial, 2024). SkillCall tells him on the call.

Chakravorty et al. 2024



Officers get their time back

~33 hours a month

50 minutes saved per person: a 40-min survey + form becomes a 10-min review, and follow-up calls are automatic. For an officer handling 40 people a month that is ~33 hours — 4 working days.

our estimate



Less money wasted

₹12.5 crore a year

If just 5 in 100 wrong matches are avoided: 2,500 people × ₹50,000 support. That is 15–34 times what SkillCall costs to run.

our estimate

WHO WINS



SC family

Right course the first time, free, in his own language — and a follow-up call until he is actually earning.



CSC operator

The case arrives pre-filled as a table; only documents and e-KYC are left to do. ₹20–40 fee per assisted profile.



District officer

Ready, checked cases in the AJAY app instead of a 40-minute survey; a demand map by block for the annual plan.



Ministry

One record from call to job; every outcome marked confirmed (employer / EPFO*) or self-reported.

BENEFITS



Social

Women and non-readers included; free, in his own language; right course the first time.



Economic

Less money spent on wrong training; more people earning — proved by follow-up, not assumed.



Governance

One shared case record for CSC, officer, partner and bank; planning data instead of paper forms.



Environment

No paper forms, fewer trips to offices, runs on the shared government cloud.



Government data

- CAG Report No. 20 of 2025 — PMKVY audit (41% placed) cag.gov.in
- PM-AJAY GIA norms & beneficiaries — PIB 2023 / 2025 pib.gov.in
- PM-AJAY budget ₹2,140 cr — Demands for Grants 2026-27 indiabudget.gov.in
- NFHS-6 (women's phone access) · Census 2011 (SC literacy) pib.gov.in



Technology

- Sarvam AI — Saaras v3, Bulbul v3, Sarvam-30B / 105B, pricing sarvam.ai
- AI4Bharat IndicConformer — open Indian speech recognition ai4bharat.iitm.ac.in
- Pipecat — open-source voice-agent framework pipecat.ai
- Bhashini (MeitY) · Exotel · PostgreSQL · FastAPI · React bhashini.gov.in



Research papers

- Chakravorty et al. 2024, J. Dev. Econ. — honest job info → +11 pts retention sciencedirect.com
- Bashingwa et al. 2021, BMJ Global Health — Kilhari at 12 lakh calls/day gh.bmi.com
- Voice of India 2026 — Indian speech-AI benchmark on real calls arxiv.org
- Medhi et al. 2011, ACM TOCHI — voice UIs for low-literacy users dl.acm.org



Standards & policy

- NCO-2015 occupation codes · NQR / NCVET job roles nqr.gov.in
- DPDP Rules 2025 — consent and data protection meitv.gov.in
- TRAI 1600-series rules for government calls (2026) traigov.in
- UIDAI-Sarvam voice AI deployment — PIB / MeitY 2025 pib.gov.in



See more

code, working prototype, demo video and the full 16-part reference file with every calculation

[GitHub repo →](#)[Live prototype →](#)[Demo video →](#)[Reference file →](#)